Predicting Credit Card Fraud

Machine Learning Assignment Report

FinGuard Analytics

1. Problem Overview

FinGuard Analytics, a fintech company, is experiencing increasing financial losses due to fraudulent credit card activity. The goal of this project is to develop a machine learning model capable of identifying fraudulent transactions with high reliability while maintaining efficiency for real-time deployment.

This is a binary classification problem where every transaction must be labelled as either:

- 0 = Legitimate transaction
- 1 = Fraudulent transaction

The single greatest challenge in this project is extreme class imbalance — fraud represents only a tiny fraction of all transactions, making standard machine learning approaches unreliable without special handling.

2. Dataset Overview

The dataset contains historical credit card transactions with the following characteristics:

Property	Detail
Total Transactions	284,807
Total Features (Columns)	31
Missing Values	None
Legitimate Transactions	284,315 (99.83%)
Fraudulent Transactions	492 (0.17%)

2.1 Feature Description

The dataset contains the following columns:

- Time — Seconds elapsed since the first transaction in the dataset
- V1 to V28 — Anonymized features already transformed by the bank using PCA for privacy protection. These represent behavioral patterns such as spending habits, location, and merchant types.
- Amount — The actual transaction amount in dollars
- Class — The target variable: 0 = Legitimate, 1 = Fraud

2.2 Class Imbalance

The dataset is severely imbalanced. Only 0.17% of transactions are fraudulent. This is the most critical challenge because a naive model that predicts every transaction as legitimate would still achieve 99.83% accuracy — yet catch zero fraud cases. This is why accuracy alone is a misleading metric for this problem.

3. Data Preprocessing

3.1 Scaling

The features V1 through V28 were already scaled by the bank. However, the Amount and Time columns were on very different scales and needed to be standardized. StandardScaler was applied to both columns to bring them into the same range as the other features. Without scaling, large values in Amount could have incorrectly dominated the model's decisions.

3.2 Train-Test Split

The dataset was divided into a training set (80%) and a testing set (20%):

- Training set: 227,845 transactions — used to teach the model
- Testing set: 56,962 transactions — used to evaluate performance on unseen data

3.3 Handling Class Imbalance with SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data to fix the class imbalance. SMOTE works by generating new synthetic fraud examples based on existing ones, rather than simply duplicating them.

	Before SMOTE	After SMOTE
Legitimate	227,451	227,451
Fraud	394	227,451

After SMOTE the model was trained on a perfectly balanced dataset, giving it equal opportunity to learn both legitimate and fraudulent transaction patterns.

4. Model Development & Evaluation

4.1 Evaluation Metrics

Given the extreme class imbalance, the following metrics were prioritised over accuracy:

- Precision — Of all transactions flagged as fraud, how many were actually fraud? Low precision means many innocent customers are wrongly flagged.
- Recall — Of all actual fraud transactions, how many did the model catch? Low recall means fraud is being missed.
- F1 Score — The harmonic mean of Precision and Recall. Provides a single balanced measure of performance.

4.2 Model 1 — Logistic Regression

Logistic Regression was used as the baseline model. It makes decisions by calculating the probability that a transaction is fraudulent based on all available features.

Metric	Legitimate	Fraud
Precision	1.00	0.06
Recall	0.97	0.93
F1 Score	0.99	0.11

While Logistic Regression achieved a high recall of 93% for fraud — meaning it caught most fraudulent transactions — its precision of just 6% is critically low. For every 100 transactions it flagged as fraud, only 6 were actually fraudulent. The remaining 94 were innocent customers wrongly accused. This makes Logistic Regression unsuitable for production deployment.

4.3 Model 2 — Random Forest

Random Forest was selected as the primary model. It works by building multiple decision trees and combining their votes to make a final prediction — making it significantly more robust and accurate than a single model.

Metric	Legitimate	Fraud
Precision	1.00	0.93
Recall	1.00	0.82
F1 Score	1.00	0.87

Random Forest delivered dramatically superior results. Precision jumped from 6% to 93%, meaning when the model flags a transaction as fraudulent it is correct 93% of the time. The F1 Score of 0.87 reflects an excellent balance between catching fraud and avoiding false accusations.

4.4 Model Comparison Summary

Model	Fraud Precision	Fraud Recall	Fraud F1 Score
Logistic Regression	6%	93%	0.11
Random Forest	93%	82%	0.87

Random Forest is the clear winner and is recommended as the production model. Although its recall is slightly lower (82% vs 93%), the massive improvement in precision makes it far more practical — it catches most fraud while causing minimal disruption to legitimate customers.

5. Principal Component Analysis (PCA)

5.1 What is PCA?

PCA (Principal Component Analysis) is a dimensionality reduction technique that compresses multiple columns into fewer components while retaining as much information as possible. It was applied here to explore whether the model could perform well with fewer features — improving speed and simplicity at the potential cost of some accuracy.

5.2 PCA Configuration

PCA was configured to retain 95% of the variance in the data, automatically determining the optimal number of components needed. This reduced the original 30 features to a smaller set of principal components.

5.3 Performance Comparison: With vs Without PCA

Metric	Without PCA	With PCA
Fraud Precision	93%	63%
Fraud Recall	82%	85%
Fraud F1 Score	0.87	0.72
Number of Features	30	Reduced

5.4 Analysis of PCA Results

Applying PCA resulted in a notable drop in precision from 93% to 63%. This means that with PCA, more innocent transactions are incorrectly flagged as fraudulent. Recall remained similar (85% vs 82%), indicating that PCA retained enough information to detect most fraud cases.

The trade-off is clear: PCA offers faster training due to fewer features but comes at a meaningful cost in precision. For fraud detection, where falsely accusing innocent customers damages trust and incurs operational costs, this trade-off is not acceptable. The full 30-feature model is recommended for deployment.

6. Feature Importance Analysis

6.1 Most Important Features

Random Forest provides an importance score for each feature, indicating how much each one contributes to fraud detection decisions. The top 10 most important features are:

Rank	Feature	Importance Score	Contribution
1	V14	41.3%	Dominant fraud signal
2	V17	13.7%	Strong signal
3	V10	10.0%	Strong signal
4	V12	7.8%	Significant signal
5	V2	5.0%	Moderate signal
6	V4	3.5%	Moderate signal
7	V11	1.5%	Weak signal
8	V8	1.4%	Weak signal
9	Amount	1.2%	Weak signal
10	V5	1.1%	Weak signal

6.2 Key Insights from Feature Importance

- V14 is overwhelmingly the most powerful fraud signal, accounting for 41.3% of all detection decisions on its own. This single feature contributes nearly half of the model's fraud-catching ability.
- The top 4 features combined (V14, V17, V10, V12) account for approximately 73% of all fraud detection power. The remaining 26 features collectively contribute only 27%.
- Transaction Amount ranked 9th with only 1.2% importance. Despite fraud transactions averaging $122.21 versus $88.29 for legitimate ones, the amount alone is a poor fraud signal. Behavioral patterns captured in the anonymized V features are far more predictive.
- Bottom features such as V22, V24, and V27 scored below 0.5% each, contributing almost nothing to detection. These could potentially be removed in future model optimization.

7. Discussion of Trade-offs

Dimension	Finding
Performance vs Interpretability	Random Forest outperforms Logistic Regression significantly but is harder to interpret. Logistic Regression is simpler to explain but performs poorly on this imbalanced problem.
Performance vs Speed (PCA)	Removing PCA preserves model accuracy (F1: 0.87 vs 0.72). While PCA speeds up training, the cost to precision is too high for a fraud detection system.
Recall vs Precision	A higher recall catches more fraud but flags more innocent transactions. Random Forest provides the best balance with 93% precision and 82% recall.
Model Complexity	Random Forest with 10 trees is fast enough for near-real-time deployment while still delivering excellent performance.

8. Conclusions & Recommendations

8.1 Final Model Recommendation

The Random Forest model trained on SMOTE-balanced data without PCA is recommended as the production fraud detection system. It delivers the best combination of precision, recall, and F1 score.

8.2 Summary of Findings

- The dataset is extremely imbalanced at 0.17% fraud rate, requiring SMOTE to ensure the model learns fraud patterns effectively.
- Random Forest significantly outperforms Logistic Regression, achieving an F1 Score of 0.87 vs 0.11 for fraud detection.
- PCA reduces model complexity but lowers precision from 93% to 63%, making it unsuitable for this use case.
- V14 is by far the most powerful fraud indicator, contributing 41.3% of detection decisions alone.
- Transaction Amount is a surprisingly weak fraud signal at only 1.2% importance.
- Accuracy is a misleading metric for imbalanced datasets — Precision, Recall, and F1 Score must be used instead.

8.3 Future Improvements

- Investigate the meaning of V14 with the bank's data science team — its dominance may reveal important fraud patterns.

- Experiment with XGBoost or LightGBM models which often outperform Random Forest on fraud detection tasks.
- Consider using a cost-sensitive learning approach that assigns higher penalties to missing fraud cases.
- Explore removing low-importance features (V22, V24, V27) to simplify the model without sacrificing performance.